



Department of Computer Science and Business Systems

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: III Semester B.Tech. CSBE

Course Code & Title: AD3491 & Fundamentals of Data Science and Analytics

Name of the Faculty member (s): Mrs.M.Jeya Sundari, AP/CSBE

Innovative Practice Description

- **Unit / Topic:** Unit 3 / Sample Distribution
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO6
- **Activity Chosen:** Learning base Teaching
- **Justification:**

Sample distribution deals with the distribution of sample statistics (like mean or variance) obtained from repeated random sampling of a population. It forms the foundation for inferential statistics. Helps students actively participate and visualize statistical concepts. Encourages deeper comprehension by connecting abstract concepts to practical applications. Experiential learning improves memory retention compared to lecture-based methods.

- **Time Allotted for the Activity:** 45 Minutes

•Details of the Implementation:

- Understand the concept of sample distribution and the central limit theorem.
- Engage → Explore → Explain → Elaborate → Evaluate
- Pose a real-world question: *"How do election polls predict results without surveying the entire population?"*
- Groups of 4-5 students simulate sampling using dice, coins, or a dataset.
- Compare sample statistics among groups. Highlight variability in sample means.
- Assign each group a real-life scenario (e.g., manufacturing, market surveys) to relate the activity to practical contexts.

- CO – PO / PSO mapping:

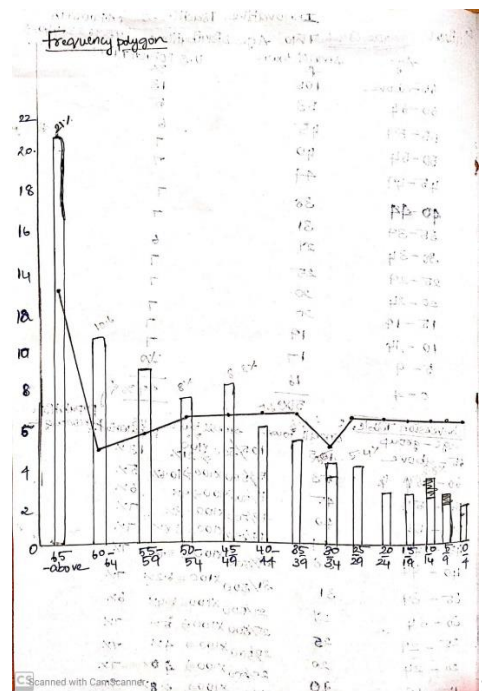
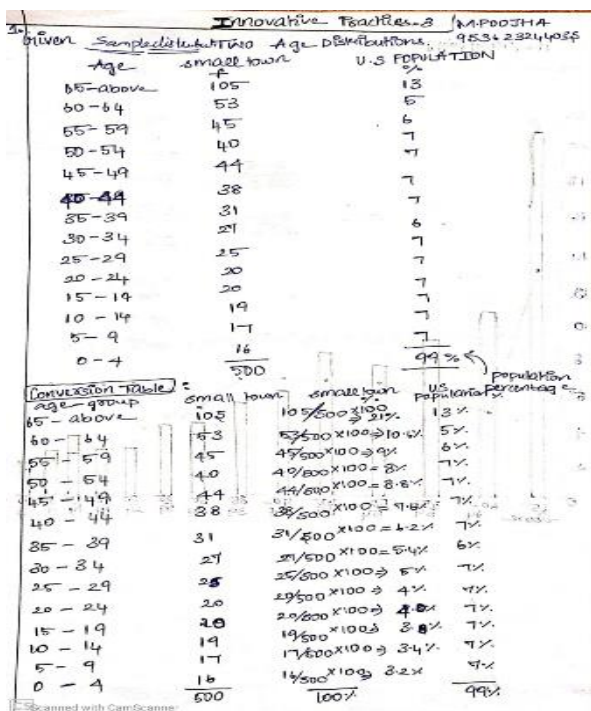
CO	PO1	PO2	PO3	PSO1	PSO2	PSO3
CO3	3	3	2	3	1	1

(1 – Low 2 – Moderate 3 – High)

- PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO1	PSO2	PSO3
	3	3	1	1	1	1
Justification for correlation	Engaging students in applying mathematical analysis, and interdisciplinary knowledge to evaluate Sample distribution .	Students to identify management-related problems, define and formulate them, collect relevant data, and propose solutions, thus enhancing their practical skills	Engaging students in investigating problems based on Sample distribution , understanding user needs, and evaluating solutions while considering constraints, thereby reinforcing their comprehensive approach to engineering challenges..	Statistical skills are essential for effectively implementing these technologies to achieve Problematic objectives in a dynamic environment.	Learning and teaching Statistical skills are crucial for aligning these technical capabilities with management objectives to drive organizational success.	Data Analytics skills are essential for effectively aligning the experiences with Analysis objectives to drive organizational growth and innovation.

- Images / Screenshot of the practice





Feedback of practice from students and other stakeholders:

- Students appreciated clear and straightforward questions, which made it easy for them to understand and respond effectively. However, some suggested avoiding jargon to ensure inclusivity.

Benefit of the practice:

- **Enhanced Understanding:** Students grasp complex concepts like the CLT through experiential learning.
- **Critical Thinking:** Encourages analysis, interpretation, and application of data.
- **Real-World Relevance:** Connects classroom learning to industry and research practices.
- **Collaborative Skills:** Builds teamwork and communication abilities essential for professional growth.

Challenges faced in implementation:

Engaging all students can be difficult, leading to low response rates. Factors such as lack of awareness, time constraints, or indifference may result in insufficient data to accurately represent the class's opinions.

References:

- <https://www.teachthought.com/learning/the-definition-of-the-flipped-classroom/#:~:text=A%20flipped%20classroom%20is%20a,the%20students%20independently%20at%20home>.
- https://en.wikipedia.org/wiki/Flipped_classroom
- [youtube.com/watch?v=BCIxikOq73Q](https://www.youtube.com/watch?v=BCIxikOq73Q)

Faculty Signature / CSBS**HOD/CSBS**